

Learning Object-Centric Barrier Functions for Contact-Rich Interaction in Clutter

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Abstract—Robots in everyday environments frequently encounter clutter, where physical contact with surrounding objects is unavoidable during both navigation and manipulation. Many safety frameworks nonetheless treat contact as failure, restricting robots to collision avoidance; model-based alternatives become intractable as the number of obstacles grows, while learned methods tie safety to the task and transfer poorly without retraining. We introduce Dense Contact Barrier Functions (DCBF), a composable, object-centric function that implicitly captures the safety constraints arising from physical interaction rather than explicitly modeling contact dynamics. Trained on sparse scenes containing only a few objects, DCBF composes to an arbitrary number of objects at runtime, yielding a single global safety filter that scales linearly in scene density. In simulation, its advantage over baselines grows with density and is largest at 40 objects, and the learned barrier transfers zero-shot to hardware without retraining.

I. INTRODUCTION

For robots deployed in unstructured spaces, safely maneuvering through and interacting with densely cluttered environments is a critical capability. It is what allows a household robot to reach into a crowded pantry, or a warehouse robot to restock shelves in tight commercial aisles. Such settings contain dense, shifting arrangements of objects, forcing the robot into physical contact in order to complete its task while having to ensure that such contact does not cause objects to tip or fall. A home robot focused on strictly avoiding collisions would be unable to complete a task beyond the first row of items in a pantry. Humans, in contrast, routinely make contact, nudging and shifting objects while implicitly understanding the interactions required to prevent spills or damage. Many existing safety frameworks treat contact itself as unsafe, making them overly restrictive in such settings. This work aims to bridge this gap, giving robots the capacity to reason about and execute safe contact.

Classical safety methods such as Hamilton–Jacobi reachability (HJR)[1], control barrier functions (CBFs)[2], and artificial potential fields (APFs)[3] have focused largely on collision avoidance and typically assume known dynamics. Learning-based approaches, such as reinforcement learning, capture complex interaction dynamics, but safety is entangled into the task policy itself, coupling safety and

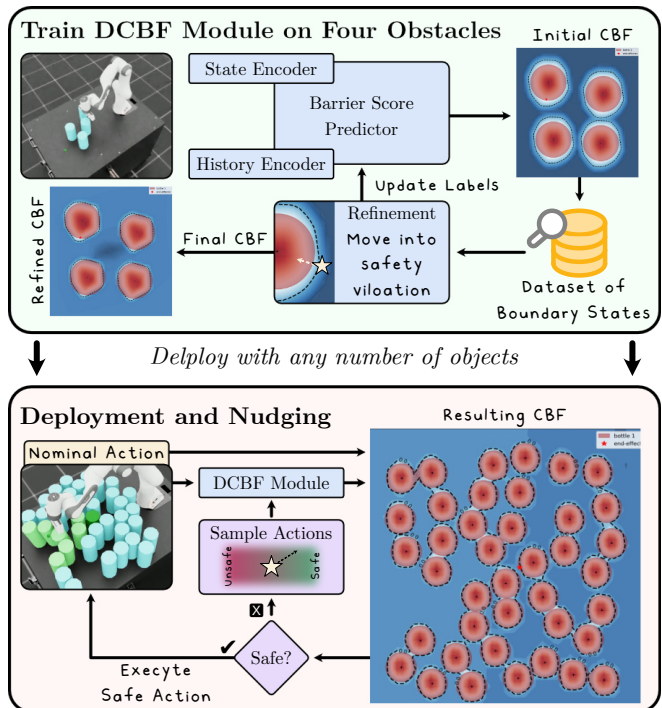


Fig. 1: Overview of the Dense Contact Barrier Function (DCBF) framework. Offline, interaction-rich trajectories from sparse scenes train an object-centric barrier, refined near its predicted boundary. At deployment, per-object scores are aggregated and a sampling-based filter returns the safe action closest to the nominal command.

performance[4], [5]. Neural CBFs instead learn safety certificates from data [6], [7], and recent work has explored compositional neural barriers for scalable safety reasoning [8]. Related efforts have also begun to extend formal safety methods beyond collision avoidance to manipulation and contact-rich interaction [9], [10], [11], but scalable safety reasoning for dense multi-object contact remains challenging.

In this work, we introduce Dense Contact Barrier Functions (DCBF, see Fig. 1), an object-centric control barrier functions that compute implicit interactions to achieve safe, scalable interaction in environments containing dense objects. The core of our framework is a composable DCBF

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that is trained offline on interactions with just a few objects. At runtime, this function is evaluated for each object in the scene, and the results are aggregated into a single safety filter. This compositional approach allows our method to scale to environments with an arbitrary number of objects and generalizes across diverse configurations without retraining. Our method avoids the intractability of explicit multi-object interaction modeling and reasoning by learning to implicitly capture the safety constraints arising from complex interactions in an object-centric, scalable manner. Our primary contributions are:

- A composable, object-centric CBF that captures multi-object interaction effects from state histories without explicit contact dynamics, treating contact as admissible rather than forbidden.
- Composition via a minimum operator that scales a barrier trained on a few objects to arbitrary clutter at runtime, linear in the number of objects.
- Validation in simulation and on real hardware, demonstrating safe contact-rich interaction in scenes far denser than those seen in training.

II. PROBLEM SETUP

We address the problem of moving through a dense group of obstacles without destabilizing them: the robot may physically contact objects, but must keep every object's tilt below a prescribed angle while remaining effective at its task. Safety is thus defined in terms of object stability rather than contact avoidance. We define the state of the robot as $\mathbf{r} \in \mathbb{R}^n$ with dynamics

$$\mathbf{r}^{t+1} = f(\mathbf{r}^t, u^t), \quad (1)$$

where $u^t \in \mathbb{R}^a$ is the control input. The environment contains N objects each with a state $\mathbf{o}_i \in \mathbb{R}^m \forall i \in [1, N]$. The safety of each object is defined by a function $q_i(\mathbf{o}_i)$, where a violation occurs if this safety cost exceeds a known threshold, i.e. $q_i(\mathbf{o}_i) \geq q_T$.

For multiple objects the dynamics, given by

$$\begin{aligned} & \{\mathbf{o}_1^{(t+1)}, \mathbf{o}_2^{(t+1)}, \dots, \mathbf{o}_N^{(t+1)}\} \\ & = F(\mathbf{r}^{(t)}, u^{(t)}, \{\mathbf{o}_1^{(t)}, \mathbf{o}_2^{(t)}, \dots, \mathbf{o}_N^{(t)}\}), \end{aligned} \quad (2)$$

are influenced by both the robot and inter-object interactions. Explicitly modeling the coupled multi-object contact dynamics becomes increasingly difficult as clutter density grows. We therefore seek a learned safety certificate that does not require online rollout of the full contact dynamics.

III. DENSE CONTACT BARRIER FUNCTIONS

Rather than explicitly learning the object dynamics, which are highly contact-dependent and often do not generalize to unseen environments, we learn a safety landscape that captures control effect implicitly. Evaluating a candidate control u at $(\mathbf{r}^t, \mathbf{o}^t)$ would require the one-step lookahead

$$h(f(\mathbf{r}^t, u), p(\mathbf{o}^t, \mathbf{r}^t, u)) = h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1}),$$

which is unavailable in practice, since the object dynamics p are unknown. Instead, we compute $\mathbf{r}^{t+1} = f(\mathbf{r}^t, u)$ using the

known robot model, and score the action via a data-driven predictor $B(\mathbf{r}^{t+1}, \mathbf{o}^t)$ serving as a proxy for the “next-step” barrier score: $B(\mathbf{r}^{t+1}, \mathbf{o}^t) \approx h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1})$. Following robust CBF formulations [12], we introduce a robustness margin σ to account for barrier-value prediction mismatch. If the prediction mismatch is bounded by the robustness margin:

$$B(\mathbf{r}^{t+1}, \mathbf{o}^t) - h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1}) \leq \sigma,$$

then enforcing the robust CBF constraints with B implies the standard CBF condition under the true dynamics:

$$B(\mathbf{r}^{t+1}, \mathbf{o}^t) - B(\mathbf{r}^t, \mathbf{o}^{t-1}) - \sigma \geq -\gamma(B(\mathbf{r}^t, \mathbf{o}^{t-1})). \quad (3)$$

By bypassing explicit dynamics modeling, B focuses on the invariant structure required for safety, rather than reconstructing complex, non-smooth contact interactions.

A. Barrier Formulation

1) *Temporal Object-State History*: We define an absolute history window of length T as $\mathbf{O}^t = [\mathbf{o}^t, \mathbf{o}^{t-1}, \dots, \mathbf{o}^{t-(T-1)}]$. This temporal context allows B to condition its safety estimate on recent motion trends rather than relying on a single instantaneous observation.

2) *Object-Centric Representation*: Inspired by [13], we adopt an object-centric representation to learn a single per-object model that can be deployed across variable-sized object sets. Let $R(a, b)$ denote a transformation that expresses state a relative to reference b . For each object i , we anchor the coordinate frame at the object's position at time $t - T$ and define

$$\begin{aligned} \mathbf{r}_i^t &= R(\mathbf{r}^t, \mathbf{o}_i^{t-T}), \\ \mathbf{O}_i^t &= [R(\mathbf{o}_i^t, \mathbf{o}_i^{t-T}), \dots, R(\mathbf{o}_i^{t-(T-1)}, \mathbf{o}_i^{t-T})]. \end{aligned}$$

In this representation, deviations from the “expected” object motion (e.g., lateral drift when pushed forward) implicitly encode the presence of object-object interactions.

3) *Inference for Multi-Object Scenes*: To certify scene-level safety, we compose the per-object barrier values via a minimum operator. For N objects, the safety for a candidate control input u is evaluated as:

$$B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) = \min_{i=1, \dots, N} B(f(\mathbf{r}_i^t, u), \mathbf{O}_i^t). \quad (4)$$

A control input is deemed admissible if $B_{\text{global}} \geq 0$, which is equivalent to requiring that all object-level barrier values are nonnegative (safe). This composition scales linearly in N and is trivially parallelizable across objects.

IV. CONTROL BARRIER TRAINING AND DEPLOYMENT

We train DCBF using interaction-rich data followed by boundary-guided refinement, and deploy the refined model as a minimally invasive safety filter.

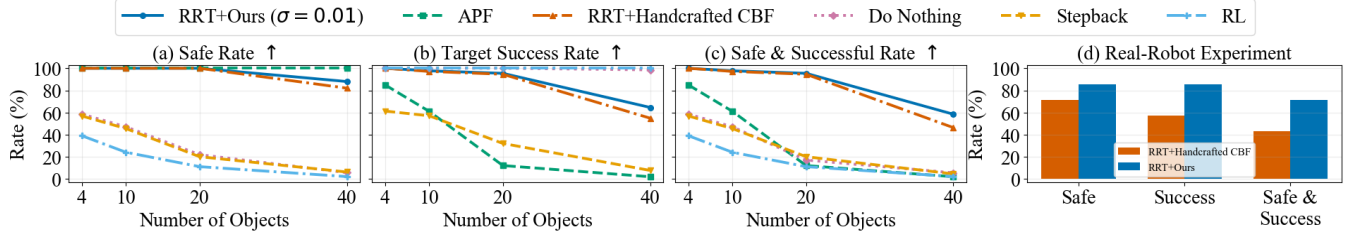


Fig. 2: Quantitative comparison of simulated and real-world experiments across different number of obstacles.

A. Data Collection

We collect interaction-rich trajectories in simulation using a suboptimal behavior policy, where interaction-rich refers to trajectories containing frequent robot–object contacts that induce object motion and generate samples near the safety boundary. Specifically, the end-effector moves toward a randomly sampled goal. During execution, if the tilt angle of any object exceeds (15°), the end-effector backtracks to the previous waypoint and samples a new random goal. This behavior policy naturally produces both safe and unsafe interactions, providing frequent samples near the safety boundary without requiring a highly tuned controller. Training scenes contain only a limited number of objects, typically far fewer than those encountered at deployment.

For each trajectory and each object i , we store transition tuples $\{(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}), (\mathbf{r}_i^{t+1}, \mathbf{O}_i^t)\}$, and assign a binary label using the ground-truth safety criterion at the resulting physical state. The labeled transitions form a safe dataset $\mathcal{D}_s$ and an unsafe dataset $\mathcal{D}_u$, with $\mathcal{D} := \mathcal{D}_s \cup \mathcal{D}_u$. For later refinement, we also store the full simulator state at each transition to enable re-simulation under alternative control inputs.

B. Initial Training

We parameterize B with a neural network comprising (i) an LSTM encoder for the temporal object history, (ii) an MLP encoder for the robot state, and (iii) a fusion head that outputs a scalar barrier value (also see Fig. 1). We obtain an initial model B_θ by minimizing the composite objective $L(\theta)$ defined as:

$$L(\theta) = \eta_s L_s(\theta) + \eta_u L_u(\theta) + \eta_d L_d(\theta), \quad (5)$$

$$L_s(\theta) = \mathbb{E}_{\mathcal{D}_s} \left[-B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \right]_+, \quad (6)$$

$$L_u(\theta) = \mathbb{E}_{\mathcal{D}_u} \left[B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \right]_+, \quad (7)$$

$$L_d(\theta) = \mathbb{E}_{\mathcal{D}_s \cup \mathcal{D}_u} \left[\begin{aligned} &(1 - \gamma) B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \\ &- B_\theta(\mathbf{r}_i^{t+1}, \mathbf{O}_i^t) + \sigma \end{aligned} \right]_+, \quad (8)$$

where $[\cdot]_+$ denotes ReLU, $\eta_s, \eta_u, \eta_d > 0$ are weighting coefficients, and $\gamma \in (0, 1)$ defines the adopted linear class- $\mathcal{K}$ function $\gamma(s) = \gamma s$. The terms L_s and L_u enforce correct sign on labeled samples, while L_d penalizes violations of the discrete-time barrier condition.

C. Boundary-Guided Refinement

This procedure is inspired by counterexample-guided approaches for neural barrier synthesis [14], [15], which iteratively identify informative or violating states to refine

a candidate certificate. In our setting, rather than using a formal verifier, we revisit near-boundary states in simulation and relabel the resulting transitions using the ground-truth safety criterion. The initial model may be conservative or violate invariance near the learned boundary because the offline dataset is biased by the behavior policy. We therefore iteratively revisit near-boundary states and relabel them through simulator rollouts.

To identify near-boundary samples, we select transitions from $\mathcal{D}$ such that $|B(\mathbf{r}_i^t, \mathbf{O}_i^{t-1})| \leq \delta$ for a tunable threshold $\delta > 0$. For each selected transition, we re-initialize the full multi-object simulator state and execute a short rollout (up to s steps) under a “safest-action” policy that maximizes the next-step global barrier value:

$$u_{\text{refine}}(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) = \arg \max_{u \in \mathbb{R}^a} B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \quad (9)$$

All transitions collected during these rollouts are annotated using the ground-truth safety criterion and merged into the training set, after which we fine-tune the model via (5). We repeat this loop until performance stabilizes.

D. Deployment

At runtime, the refined barrier is used as a safety filter that minimally perturbs a nominal control input u_{norm} to ensure safety. We use sampling-based action selection as follows:

- 1) Evaluate the nominal control. If $B_{\text{global}}^{u_{\text{norm}}}(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \geq 0$, execute u_{norm} .
- 2) Otherwise, sample a candidate set $\mathbb{U}$ around u_{norm} (e.g., Gaussian perturbations in control space).
- 3) Form the safe subset $\mathbb{U}_s = \{u \in \mathbb{U} \mid B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \geq 0\}$ and execute $u_{\text{filtered}} = \arg \min_{u \in \mathbb{U}_s} \|u - u_{\text{norm}}\|$.

V. EXPERIMENTS AND RESULTS

We evaluate our method in simulation and on a real-world hardware setup, demonstrating safe, scalable contact-rich interaction in cluttered environments where physical contact is unavoidable.

Experiment Setup: Experiments are conducted in a custom IsaacLab environment with a Franka Emika Panda arm and cylindrical bottles (see Fig. 1 examples). In the real-world, we leverage a Fairino FR5 robot and 50g cardboard cylinders with a diameter of 7cm and height of 23cm (see Fig. 3). The robot must reach a randomly sampled planar target while ensuring that no bottle exceeds a tilt angle of 15° . The end-effector is constrained to planar (x, y) motion and carries a small cylinder through the clutter. The action space consists of 2D waypoint increments with a maximum

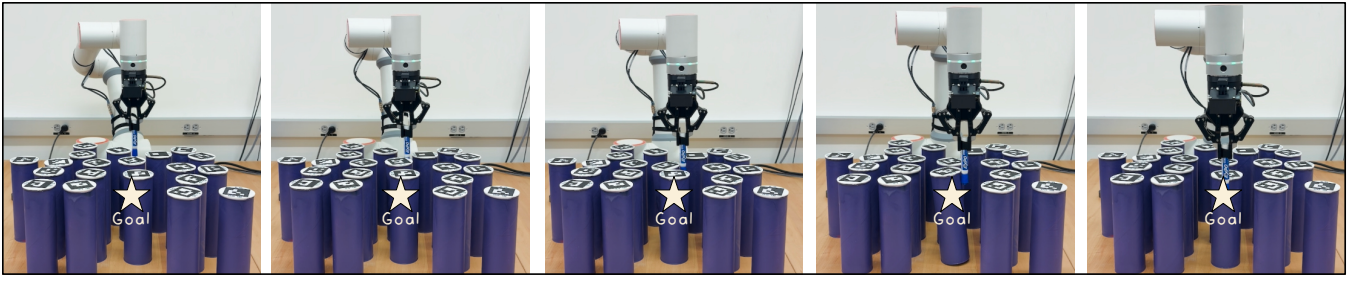


Fig. 3: Demonstration of DCBF on a real robot platform, navigating through dense clutter. The robot is nudging the obstacles step two and three without tipping them over, ultimately reaching its goal in step 5.

step size of 1 cm. Each object state is $\mathbf{o}^t = (x^t, y^t, z^t, \theta^t)$, where (x^t, y^t, z^t) denotes the bottle’s Cartesian position, and θ^t is the tilting angle. Relative transforms are computed by subtracting (x, y) coordinates in the horizontal plane.

Barrier Training: Our DCBF is trained only in simulation using environments that contain only 4 bottles. The initial dataset comprises 1,200 trajectories collected with a naive backtracking policy: the robot moves toward a sampled goal and, whenever any object’s tilt approaches the safety threshold, returns to the previous waypoint and resamples a new goal. We then discard samples in free space, focusing the dataset on contact-rich interactions.

Simulation Evaluation: We first evaluate the proposed models in simulation and assess generalization by deploying it in scenes with 4, 10, 20, and 40 objects.

We compare against five baselines: (i) **Do Nothing**, moves directly toward the goal in a straight line; (ii) **Back-stepping**, the reactive data-collection policy that retreats when any object’s tilt approaches 14° ; (iii) **Artificial Potential Fields (APF)**, a repulsive-field method with parameters well-tuned to allow limited contact without excessive conservativeness; (iv) **RRT-CBF**: a classical model-based baseline that uses RRT to generate a nominal path toward a fixed goal and a handcrafted geometric CBF as an online safety filter; (v) **RL-Safety**: a reinforcement-learning policy trained end-to-end with a task reward encouraging progress toward the goal and a safety reward penalizing unsafe object states; When applying our safety filter, Do Nothing serves as the nominal controller.

Quantitatively, Fig. 2(c) reports the safe-and-successful rate, while (a) and (b) separately report safety and task success. DCBF is the only method that retains substantial safe-and-successful performance at 40 objects, where APF, Handcrafted CBF, and the reactive baselines all fall below 20%. At lower densities it matches the strongest baseline while remaining near-saturated. The purely conservative baselines retain perfect safety at high density but achieve it by stalling, reaching the goal in fewer than 10% of trials; DCBF instead accepts a modest reduction in safety rate to preserve task completion, a trade-off governed by the robustness margin σ .

Hardware Evaluation: Furthermore, we evaluate the same DCBF trained in simulation on real hardware. We evaluate the proposed model zero-shot on hardware after training entirely in simulation. Twenty objects are placed at

random with inter-object spacings of approximately 1–3 cm, matching the density used in the simulated experiments. Object poses are estimated from ArUco markers mounted on top of each object and tracked by two Orbbec Femto Mega cameras. We define safety as the absence of knocked-over objects rather than enforcing a strict tilt-angle threshold and adopt a binary safety metric based on whether an object falls over. Figure 2(d) reports the hardware results. The observed performance mirrors the trends in simulation, underlining the generalization capabilities of our DCBF method.

VI. CONCLUSION

Our results suggest that object-centric learned barriers provide a scalable runtime safety layer for contact-rich manipulation. The advantage over baselines grows with scene density and is largest at 40 objects, consistent with per-object composition being what enables scaling, and the barrier transfers zero-shot to hardware without retraining. Two limitations remain: the learned barrier is not globally verified over the continuous state–action space, and our global certificate takes a minimum over per-object values, so its guarantees depend on whether finite histories capture the relevant coupled interactions. Future work will broaden the experimental evaluation and extend boundary-guided refinement with formal counterexample generation.

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